

ML MINI PROJECT (ORANGE PROBLEM)

TITLE: Resume–Job Description Matcher

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Overview

An AI-powered system that evaluates how well a resume aligns with a job description using semantic understanding and keyword analysis. Built with Streamlit, it offers real-time matching, category prediction, and interpretability through detailed keyword insights.

Key Highlights

- Fine-tuned SentenceTransformer (MiniLM-L6-v2): Learns contextual relationships between resumes and JDs using *CosineSimilarityLoss*.
- Hybrid Feature Fusion: Combines semantic embeddings with TF-IDF and CountVectorizer features for both deep and lexical matching.
- Explainable Output: Displays keyword overlap, missing JD terms, coverage %, and alignment insights.
- Interactive Interface: Upload resumes and view instant results within a clean Streamlit dashboard.

Core Components

| Component          | Model / Method           | Purpose                     |
|--------------------|--------------------------|-----------------------------|
| Semantic Encoding  | SentenceTransformer      | Contextual text embeddings  |
| Fine-tuning Loss   | CosineSimilarityLoss     | Aligns resume–JD pairs      |
| Classifier         | RandomForest             | Predicts resume category    |
| Feature Extraction | TF-IDF + CountVectorizer | Captures keyword relevance  |
| Similarity Metric  | Cosine Similarity        | Measures semantic closeness |

Outcome

Bridges traditional keyword-based filters and AI-driven semantic recruitment. Offers an explainable, context-aware resume screening approach suitable for ATS systems and candidate-job fit analytics.

